The square of the Riemann zeta function gives rise to critical multiplicative chaos

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Goals of the talk:

- To (roughly) explain what multiplicative chaos is
- Will end up focusing on the case of *critical* multiplicative chaos
- To (roughly) state a conjecture of Saksman and Webb that relates powers of $|\zeta(s)|$ to multiplicative chaos
- To say a little about work of H.–Saksman–Webb, proving the critical case of their conjecture

What is multiplicative chaos?

Multiplicative chaos is a class of probabilistic objects first studied by Kahane in 1985.

Idea: form a random measure (i.e. a random weighting) by integrating test functions against the exponential of some collection of random variables $(X(h))_{h\in\mathcal{H}}$.

For g a test function, we can look at

$$\int g(h)e^{\gamma X(h)}dh,$$

where $\gamma > 0$ is a real parameter.

One needs to make assumptions on $X(h)$ in order for the random measure to be interesting. It turns out one gets something very interesting if the $X(h)$ are:

- Gaussian random variables;
- with mean zero $\mathbb{E}X(h) = 0$, and the same (or similar) finite non-zero variance $\mathbb{E}X(h)^2$ for all h;
  (This condition implies that the *average* mass $\mathbb{E}e^{\gamma X(h)} = e^{(\gamma^2/2)\mathbb{E}X(h)^2}$ assigned to each point h is roughly the same.)
- and the covariance $\mathbb{E}X(h)X(h')$ (i.e. the dependence between $X(h)$ and $X(h')$) decays *logarithmically* as $|h - h'|$ increases.

Note in particular that the $X(h)$ will *not* be independent here.

Connection with number theory

Somewhat surprisingly, it turns out that multiplicative chaos is closely connected to number theory, especially the value distribution of L-functions and of sums of multiplicative functions (e.g. Dirichlet characters).

Key Example (in this talk!): shifts of the Riemann zeta function

We study the behaviour of

$$\int_{-1/2}^{1/2} g(h)|\zeta(1/2+it+ih)|^{\gamma}dh,$$

as $T \leq t \leq 2T$ varies.

Here $g(h)$ will be a suitably nice, fixed test function (e.g. a continuous function, or a step function). *The case where $g(h) \equiv 1$ already captures the key ideas.*

- By the Selberg central limit theorem, $\log|\zeta(1/2+it)|$ has a roughly Gaussian distribution as t varies.
- As briefly hinted yesterday (branching random walk), the values $\log|\zeta(1/2+it_1)|, \log|\zeta(1/2+it_2)|$ will be correlated (i.e. not statistically independent) when $|t_1-t_2|\leq 1$, say.
- So looking at $\int_{-1/2}^{1/2} g(h)|\zeta(1/2+it+ih)|^\gamma dh$ for varying t *might* be related to multiplicative chaos.

Goal

Our goal is a statement of the shape: there is a normalising factor $N = N(T, \gamma)$ such that, for any nice test function $g(h)$, the quantity

$$\frac{1}{N} \int_{-1/2}^{1/2} g(h) |\zeta(1/2 + it + ih)|^{\gamma} dh, \qquad T \leq t \leq 2T \text{ uniformly random}$$

converges in distribution to some (interesting) limiting random variable $C(g)$ as $T \to \infty$.

Some issues to ponder, I

Compare with the Selberg central limit theorem for $\log|\zeta(1/2+it)|$, with $T \le t \le 2T$ uniformly random:

- $\int_{-1/2}^{1/2} g(h)|\zeta(1/2+it+ih)|^{\gamma} dh$ involves "many" different values of zeta at the same time (in fact about $\asymp \log T$ "genuinely different" values).
- Here $\log|\zeta(1/2+it+ih)|$ appears in the exponent, so we couldn't even allow additive errors of size $O(1)$ when approximating it, if we want to prove convergence in distribution.

Some issues to ponder, II

What should the normalising factor $N = N(T, \gamma)$ be?

If we average over t, we see

$$\begin{aligned} & \frac{1}{T}\int_T^{2T}\int_{-1/2}^{1/2} g(h)|\zeta(1/2+it+ih)|^\gamma dhdt \\ = & \int_{-1/2}^{1/2} g(h)\left(\frac{1}{T}\int_T^{2T}|\zeta(1/2+it+ih)|^\gamma dt\right)dh, \end{aligned}$$

where we expect/know that the moment on the inside grows like $\log^{\gamma^2/4} T$ as $T \to \infty$.

Does this mean we should take $N(T, \gamma) = \log^{\gamma^2/4} T$?

Key properties of multiplicative chaos

- As γ increases, $\mathbb{E}e^{\gamma X(h)} = e^{(\gamma^2/2)\mathbb{E}X(h)^2}$ increases, and $\int g(h)e^{\gamma X(h)}dh$ is dominated more and more by very large values of $X(h)$.
- There is a *critical value* γ_c of γ at which, with very high probability, one no longer finds any values of h for which $X(h)$ is large enough to overcome $\mathbb{E}e^{\gamma X(h)}$.
- When $\gamma < \gamma_c$, one see non-trivial behaviour after rescaling $\int g(h)e^{\gamma X(h)}dh$ by $e^{(\gamma^2/2)\mathbb{E}X(h)^2}$.
- When $\gamma = \gamma_c$, one sees non-trivial behaviour after rescaling by

$$\frac{e^{(\gamma^2/2)\mathbb{E}X(h)^2}}{\sqrt{\mathbb{E}X(h)^2}}.$$

- In the shifted zeta example, it turns out that the critical exponent $\gamma_c = 2$, and $\sqrt{\mathbb{E}X(h)^2} \asymp \sqrt{\log\log T}$.

The bad side: Critical multiplicative chaos is delicate to work with and analyse.

The good side: The exponent 2 gives us squares, so at least we can expand out in some of the calculations!
And we get interesting and unexpected results, saving something like a factor $\sqrt{\log \log T}$ more than naively expected.

(Approximately) Saksman and Webb's conjecture, 2016

For any fixed $\gamma \leq 2$, and any (say) step function $g(h)$, the quantity

$$\frac{1}{N}\int_{-1/2}^{1/2} g(h)|\zeta(1/2+it+ih)|^{\gamma} dh, \qquad T \leq t \leq 2T \text{ uniformly random}$$

will *converge in distribution* as $T \to \infty$, if we take

- $N = N(T,\gamma) = \log^{\gamma^2/4} T$ for $\gamma < 2$;
- $N = N(T,\gamma) = \frac{\log T}{\sqrt{\log\log T}}$ for $\gamma = 2$.

The limiting distribution $C(g)$ is given by the integral of $g(h)$ against (a smooth perturbation of) a "well understood" Gaussian multiplicative chaos measure.

Theorem 1 (H.–Saksman–Webb, in preparation)

Saksman and Webb's conjecture is true in the critical case $\gamma = 2$.

We find in particular that

$$\frac{\sqrt{\log\log T}}{\log T}\int_{-1/2}^{1/2} |\zeta(1/2+it+ih)|^2 dh, \qquad T \le t \le 2T \text{ uniformly random}$$

converges in distribution (to a limit distribution arising from the total mass of critical chaos) as $T \to \infty$.

Ideas from the proof

How can we prove convergence in distribution?

A possible approach is the **method of moments**:

- if $(X_n)_{n=1}^{\infty}$ is a sequence of random variables;
- if the target distribution X is "nice enough";
- and if we can show that $\mathbb{E}X_n^r \to \mathbb{E}X^r$ as $n \to \infty$, for each fixed $r \in \mathbb{N}$;

then it follows that X_n converges in distribution to X.

For example, this is how the Selberg central limit theorem for $\log|\zeta(1/2+it)|$ is proved.

But there are significant problems with applying this to $\frac{\sqrt{\log\log T}}{\log T}\int_{-1/2}^{1/2}|\zeta(1/2+it+ih)|^2dh$.

- The target limit distribution is not "nice enough", in fact all of its moments (with $r \geq 1$) are infinite!
- Computing arbitrarily high moments of $\frac{\sqrt{\log\log T}}{\log T}\int_{-1/2}^{1/2}|\zeta(1/2+it+ih)|^2dh$ would require computing arbitrarily high moments of zeta, which we cannot do (achieving asymptotics) even assuming the Riemann Hypothesis.

Strategy (the number theory part):

To proceed, we will show that *for most* $T \leq t \leq 2T$ (and up to scaling factors):

1. we can replace $\int_{-1/2}^{1/2} |\zeta(1/2 + it + ih)|^2 dh$ by $\int_{-1/2}^{1/2} |\mathcal{D}(1/2 + it + ih)|^2 dh$, where $\mathcal{D}(s)$ is a "nice" Dirichlet polynomial *with length* $T^{1/2-o(1)}$;
2. we can replace $\mathcal{D}(s)$ by a Dirichlet polynomial $\mathcal{P}(s)$, supported only on smooth numbers, and *with length* $T^{o(1)}$.

Notice that (2) gives us hope of ultimately being able to compute high moments of the integral.

Strategy (the probabilistic part):

We show:

3. we can replace our integral by $\int_{-1/2}^{1/2} B(t+h)|\mathcal{P}(1/2+it+ih)|^2 dh$, where $B(\cdot)$ is a suitable (smooth approximation to a) *barrier* that restricts the growth of $\mathcal{P}$ *and its subsums*;
4. the moments of $\int_{-1/2}^{1/2} B(t+h)|\mathcal{P}(1/2+it+ih)|^2 dh$ don't grow too fast, *and* closely match the random model where the $(p^{-it})_{p \text{ prime}}$ are replaced by independent Steinhaus random variables $f(p)$;
5. the random model converges in distribution to the desired limit.

(5) relies on work of Saksman and Webb from 2016, together with the fact that $\mathcal{P}(s)$ behaves a lot like an Euler product.

Replacing zeta by $\mathcal{D}(s)$

By the approximate functional equation, we know

$$\begin{aligned}
& \int_{-1/2}^{1/2} |\zeta(1/2+it+ih)|^2 dh \\
\approx & \int_{-1/2}^{1/2} \Big| \sum_{n \le \sqrt{\frac{T}{2\pi}}} \frac{1}{n^{1/2+it+ih}} + \chi(1/2+it+ih) \sum_{n \le \sqrt{\frac{T}{2\pi}}} \frac{1}{n^{1/2-it-ih}} \Big|^2 dh \\
= & \; 2 \int_{-1/2}^{1/2} \Big| \sum_{n \le \sqrt{\frac{T}{2\pi}}} \frac{1}{n^{1/2+it+ih}} \Big|^2 dh + \text{cross terms involving } \chi.
\end{aligned}$$

We want to show that for most $T \le t \le 2T$:

- the cross terms are negligible (compared with $\frac{\log T}{\sqrt{\log\log T}}$);
- we can replace $\sum_{n \le \sqrt{\frac{T}{2\pi}}} \frac{1}{n^{1/2+it+ih}}$ by a "nice" Dirichlet polynomial $\mathcal{D}(1/2+it+ih)$, again up to an error that is $o(\frac{\log T}{\sqrt{\log\log T}})$.

We can show the cross terms are usually negligible by shifting our integral to the left in the complex plane (e.g. to the $\Re(s) = 1/4$ line), and then integrating over $T \leq t \leq 2T$ and applying the classical mean value theorem for Dirichlet polynomials.

It is crucial that we have an integral over h that we can shift.

It turns out that we can usually replace $\sum_{n \le \sqrt{\frac{T}{2\pi}}} \frac{1}{n^{1/2+it+ih}}$ by $\mathcal{D}(1/2+it+ih)$, where we take

$$\mathcal{D}(s) := \mathcal{P}(s) \cdot \mathcal{R}(s),$$

$$\mathcal{P}(s) := \sum_{n \le X, n \text{ is } P \text{ smooth}} \frac{1}{n^s},$$

$$\mathcal{R}(s) := \sum_{n \le T^{1/2-o(1)}, n \text{ is } P \text{ rough}} \frac{1}{n^s},$$

for suitable parameters P and X (both $T^{o(1)}$).

Why? Use the mean value theorem for Dirichlet polynomials. We only need to beat $\frac{\log T}{\sqrt{\log \log T}}$, and "most" n involved in $\sum_{n \le \sqrt{\frac{T}{2\pi}}} \frac{1}{n^{1/2+it+ih}}$ appear in $\mathcal{P}(s) \cdot \mathcal{R}(s)$.

Notice $\mathcal{D}(s)$ has total length $X \cdot T^{1/2-o(1)} = T^{1/2-o(1)}$.

Replacing $\mathcal{D}(s)$ by $\mathcal{P}(s)$

Next we want to show that provided P isn't too small, for most $T \leq t \leq 2T$ we will have

$$\int_{-1/2}^{1/2} |\mathcal{D}(1/2+it+ih)|^2 dh \approx \frac{\log(\sqrt{T})}{\log P} \int_{-1/2}^{1/2} |\mathcal{P}(1/2+it+ih)|^2 dh.$$

In other words, the contribution from the rough part $\mathcal{R}(1/2+it+ih)$ is usually just a *constant multiplier* $\frac{\log(\sqrt{T})}{\log P}$.

Why should this be so?
$\mathcal{R}(1/2+it+ih)$ depends only on the p^{-it-ih} for large primes $p > P$. These terms will rotate quite rapidly as h varies, so we might expect/hope that their contribution to the h integral will usually average out to some constant.

Rough idea for how to prove this: Compute the "variance"

$$\frac{1}{T}\int_T^{2T} \Big| \int_{-\frac{1}{2}}^{\frac{1}{2}} |\mathcal{D}(1/2+it+ih)|^2 dh - \frac{\log(\sqrt{T})}{\log P} \int_{-\frac{1}{2}}^{\frac{1}{2}} |\mathcal{P}(1/2+it+ih)|^2 dh \Big|^2 dt,$$

and prove it is rather smaller than $(\frac{\log T}{\sqrt{\log\log T}})^2$.

This computation has similar complexity to a fourth moment calculation, which we can hope to perform since $\mathcal{D}(s)$ has length $T^{1/2-o(1)}$.

Actually this won't work, the variance will not be small. But it will be if one first inserts a suitable barrier into the inner integrals (to control "blow-up on the diagonal").